MMDU MULLANA OFFICIAL SYLLABUS • BCSE-012 • UNIT 3

# Unit 3: HTML5 & Modern Web Application APIs (Semantics, Canvas, Storage & Media)

**Subject:** Fundamentals of Web Technologies (BCSE-012) • **Exam Weightage:** 24 - 28 Marks • **Study Time:** 40 mins

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## 1. Evolution & Advantages of HTML5 over HTML4

### 1. The Paradigm Shift: HTML4 Limitations vs. HTML5 Solutions

Ratified as an official W3C Recommendation in October 2014, **HTML5** modernized web development from a simple text formatting language into a robust application execution runtime.

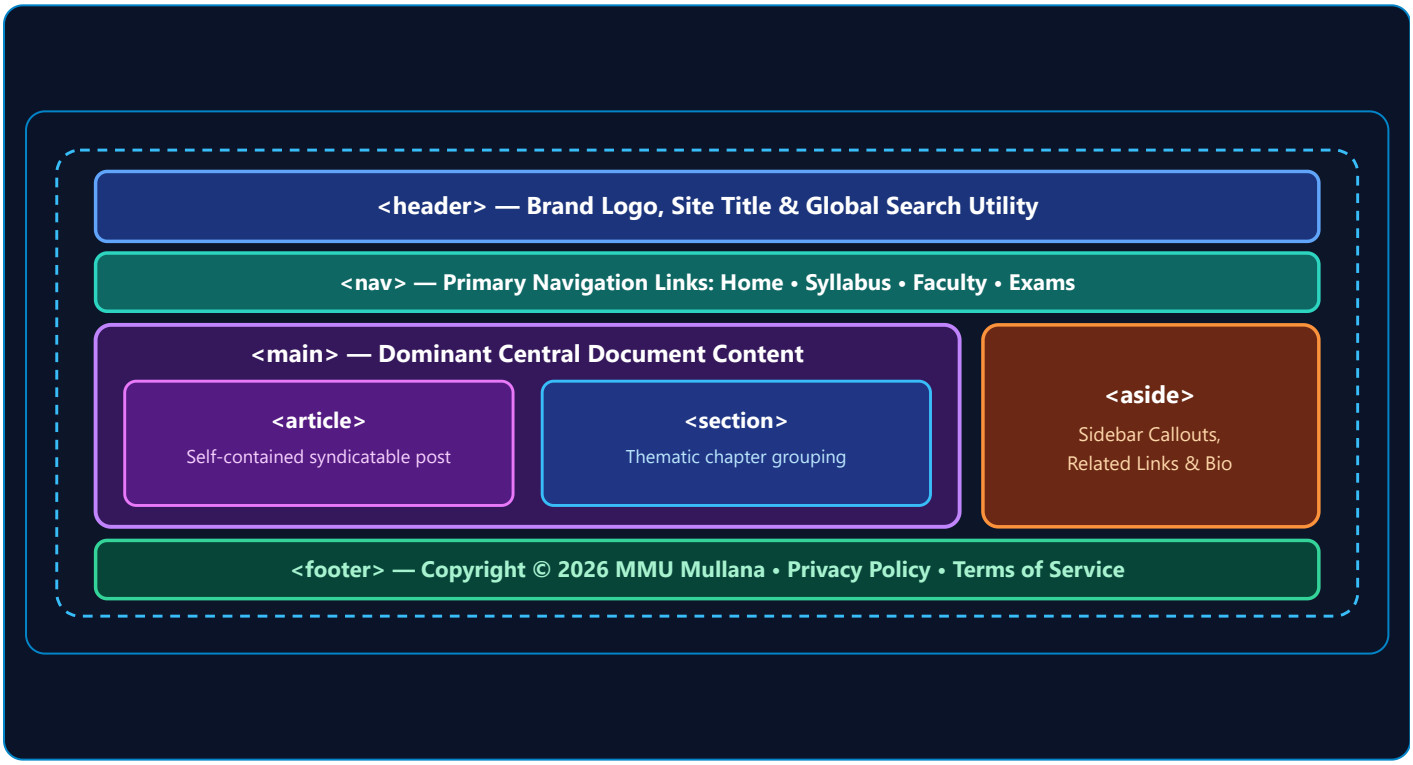
| HTML4 Limitation | Problem Encountered | HTML5 Modern Solution |
| --- | --- | --- |
| **Lack of Semantic Structure** | Heavy reliance on meaningless and generic "div-soup". Screen readers could not parse document meaning. | Native structural semantic elements ( , , , , , , ). |
| **Plugin Vulnerabilities & Crashes** | Video and audio playback required proprietary browser plugins (Adobe Flash, Silverlight), causing crashes and severe security holes. | Native hardware-accelerated and tags with multi-codec fallbacks and subtitle tracks. |
| **Inadequate Client-Side Storage** | Depended solely on 4KB HTTP Cookies transmitted across the network with every HTTP request header, causing network overhead. | Web Storage API ( `localStorage` and `sessionStorage` ) offering 5MB–10MB isolated client-side storage. |
| **No Native Graphics Rendering** | Drawing diagrams, charts, or game animations required complex Flash applets. | High-performance scriptable 2D raster drawing via the API and native scalable vector graphics ( ). |

| HTML4 Limitation | Problem Encountered | HTML5 Modern Solution |
| --- | --- | --- |
| **Poor Mobile & Error-Handling Discipline** | HTML4 lacked standardized rules for malformed markup, causing browsers to implement inconsistent parsing workarounds. | Strict standardized parsing algorithms, mobile-first responsive design, and offline capability via Service Workers. |

## 2. HTML5 Structural Semantic Elements Layout Blueprint

### Structural Semantic Architecture of a Modern HTML5 Webpage

HTML5 introduces structural semantic elements that explicitly describe their meaning to both browser rendering engines and screen readers:




**Detailed Definitions of Semantic Tags:**

- : Represents introductory content, branding logos, navigational search utilities, or author headings.
- : Encloses the major primary navigation blocks containing links to other pages or sections.
-

: Contains the central dominant content unique to that document; must not be duplicated inside headers or footers.

- : A complete, self-contained composition that is independently distributable or reusable (e.g. blog post, news story).
- : Represents a standalone thematic grouping of content, typically with its own heading (e.g. chapters, tabs).
- : Content tangentially related to the main content (e.g. sidebars, callout boxes, related article links).
- : Houses author information, copyright notices, terms of service, and back-to-top links.

## 3. Native Multimedia & Modern HTML5 Form Controls

### 1. Native Multimedia (

HTML5 eliminates third-party plugins by embedding media directly in the DOM:

```
<!-- Native HTML5 Video Element with Multi-Codec Fallback & Subtitles -->
<video width="640" height="360" controls poster="poster_image.jpg" preload="metadata">
  <source src="lecture.mp4" type="video/mp4">
  <source src="lecture.webm" type="video/webm">
  <track src="subtitles_en.vtt" kind="subtitles" srclang="en" label="English">
  Your browser does not support native HTML5 video playback.
</video>

<!-- Native HTML5 Audio Element -->
<audio controls preload="auto">
  <source src="podcast.mp3" type="audio/mpeg">
  <source src="podcast.ogg" type="audio/ogg">
  Your browser does not support the audio element.
</audio>
```

### 2. Modern HTML5 Form Inputs & Attributes

HTML5 modernized forms with built-in client-side validation without requiring custom JavaScript:

```
<form action="/submit_admission" method="POST">
  <!-- Email validation with regex check -->
  <label for="email">University Email:</label>
  <input type="email" id="email" name="email" placeholder="student@mmumullana.org" required>

  <!-- Numeric bounds -->
  <label for="age">Age (17-30):</label>
  <input type="number" id="age" name="age" min="17" max="30" required>

  <!-- Slider input -->
  <label for="scale">Skill Level (1-10):</label>
  <input type="range" id="scale" name="skill" min="1" max="10" value="7">

  <!-- Native Date Picker -->
  <label for="dob">Date of Birth:</label>
  <input type="date" id="dob" name="dob">

  <!-- Datalist Autocomplete -->
  <label for="course">Choose Branch:</label>
  <input list="branchList" id="course" name="branch">
  <datalist id="branchList">
    <option value="Computer Science & Engineering">
    <option value="Information Technology">
    <option value="Mechanical Engineering">
  </datalist>

  <button type="submit">Submit Application</button>
</form>
```

## 4. Canvas 2D Graphics API & Web Storage (Cookies vs LocalStorage)

### 1. The HTML5

The  tag provides an immediate-mode resolution-dependent bitmap surface scriptable via JavaScript:

```
<canvas id="barChart" width="400" height="180" style="border:1px solid #cbd5e1;"></canvas>
<script>
  const canvas = document.getElementById("barChart");
  const ctx = canvas.getContext("2d");

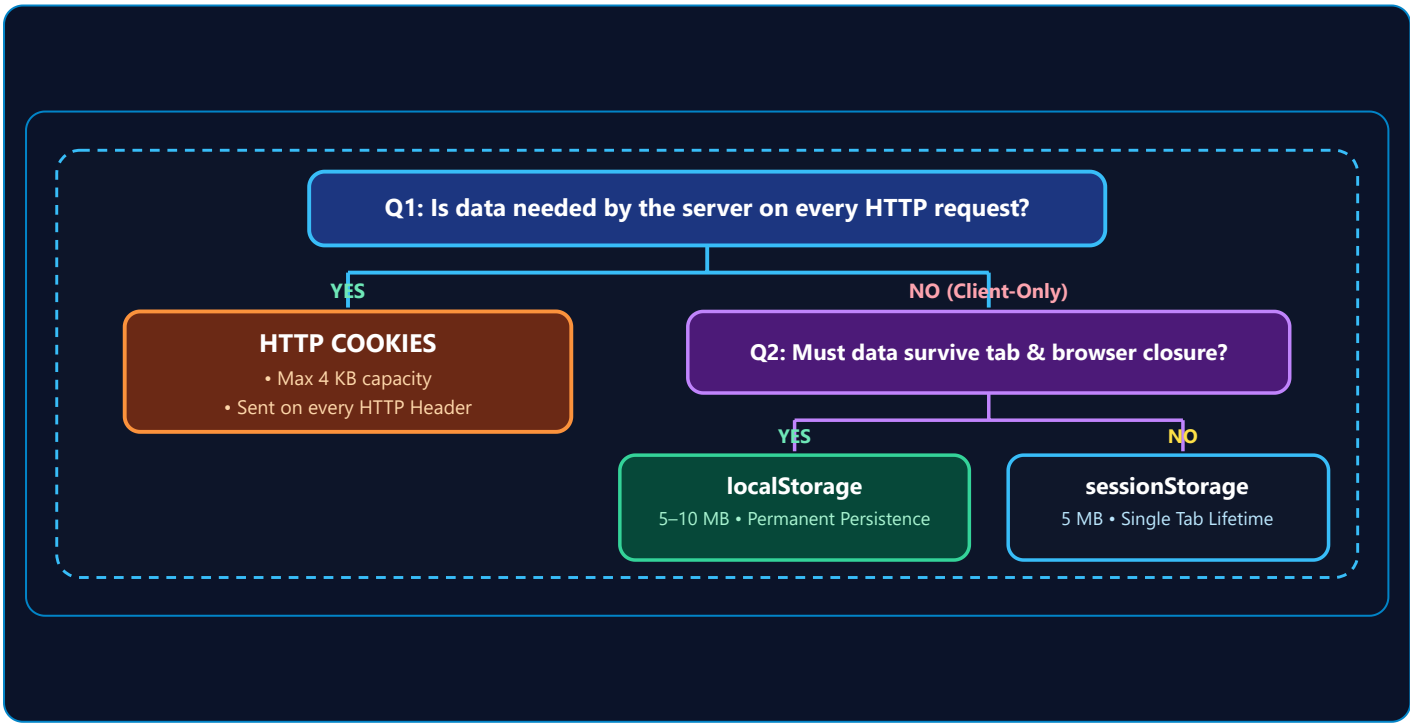
  // 1. Draw Rectangular Bar
  ctx.fillStyle = "#0284c7";
  ctx.fillRect(40, 50, 50, 100);

  // 2. Draw Second Bar
  ctx.fillStyle = "#10b981";
  ctx.fillRect(120, 30, 50, 120);

  // 3. Draw Circle
  ctx.beginPath();
  ctx.arc(280, 90, 40, 0, 2 * Math.PI);
  ctx.fillStyle = "#f59e0b";
  ctx.fill();
  ctx.stroke();

  // 4. Render Text Label
  ctx.font = "bold 13px Segoe UI, sans-serif";
  ctx.fillStyle = "#0f172a";
  ctx.fillText("Performance Chart", 40, 25);
</script>
```

## 2. Client-Side Web Storage: Cookies vs. LocalStorage vs. SessionStorage



| Dimension | HTTP Cookies | LocalStorage (HTML5) | SessionStorage (HTML5) |
|---|---|---|---|
| **Storage Capacity** | Strictly ~4 Kilobytes | ~5 to 10 Megabytes | ~5 Megabytes |
| **Data Lifetime** | Configurable via `Expires` or `Max-Age` | **Permanent** (persists after browser restart) | **Session-Only** (cleared when browser tab closes) |
| **Network Transmission** | Sent automatically on every HTTP header | **Never sent to server** (Client-only) | **Never sent to server** (Client-only) |
| **Access Scope** | Domain & path scoped | Origin scoped (shared across all tabs) | Tab scoped (isolated to single window) |
| **JavaScript API** | Complex string manipulation ( `document.cookie` ) | `setItem()` , `getItem()` , `removeItem()` | `setItem()` , `getItem()` , `removeItem()` |

## 5. University Solved Exam Questions (With Schematics)

### University Examination Solved Questions (Unit III)

#### 📘 Section A (1 Mark): LocalStorage vs SessionStorage Lifetimes

**University Question (1 Mark):**
Differentiate the persistence lifetime of localStorage vs. sessionStorage as illustrated in Figure Q3.

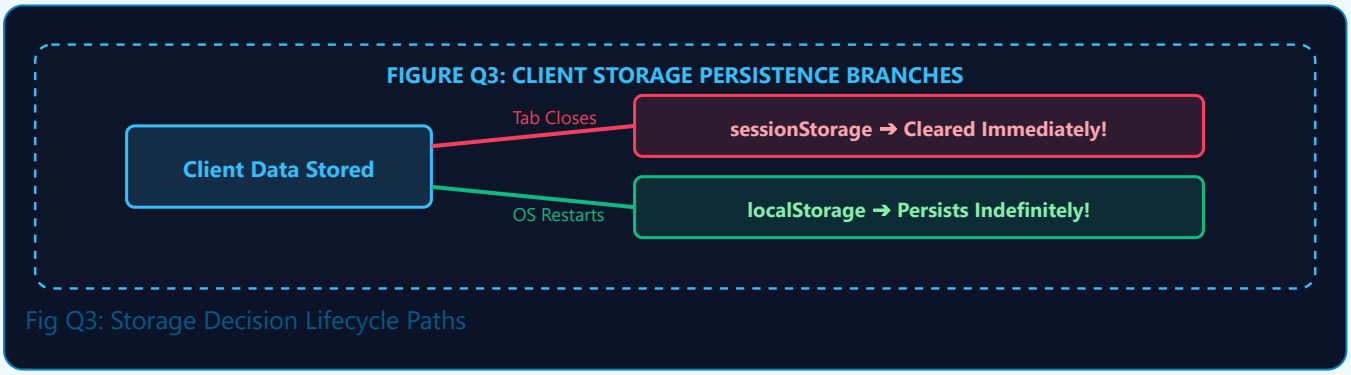


Fig Q3: Storage Decision Lifecycle Paths

**Step-by-Step Solution:**

- **`localStorage` :** Persists across browser sessions and computer reboots until explicitly cleared by the user or script ( `localStorage.clear()` ).
- **`sessionStorage` :** Confined to the lifespan of the specific browser tab; automatically purged upon tab closure.

🎯 `sessionStorage` = Tab Lifetime (Temporary) | `localStorage` = Permanent (Across Restarts).

## 📘 Section D (6 Marks): Dynamic Canvas Bar Chart Implementation

**University Question (6 Marks):**

Develop complete JavaScript and HTML5 code to draw a dynamic comparison bar chart with three subject scores on a

as shown in Figure Q15.

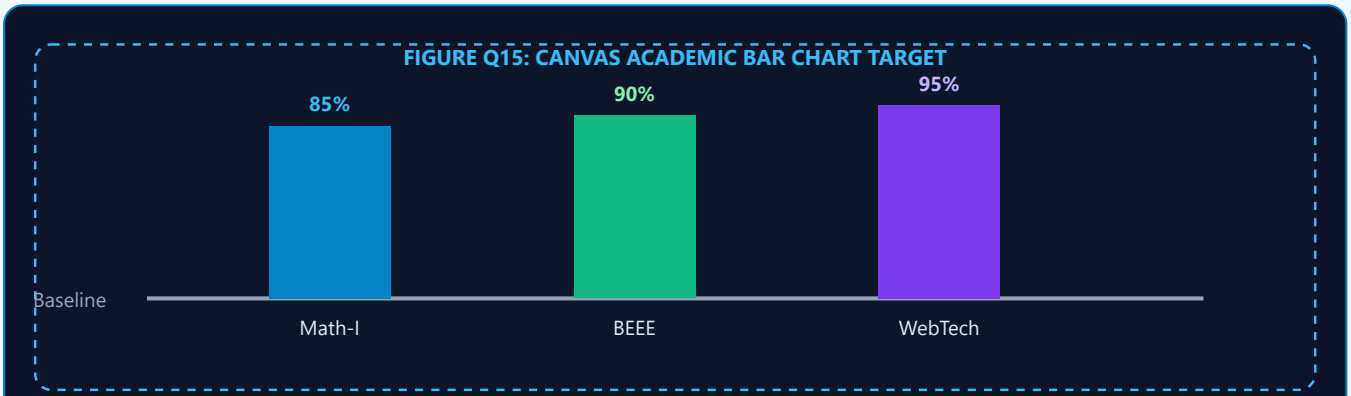



Fig Q15: Procedural Canvas Raster Visualization Target

**Step-by-Step Solution:**

```
<canvas id="chart" width="450" height="200" style="background:#f8fafc; border:1px solid
#cbd5e1;"></canvas>
<script>
  const canvas = document.getElementById("chart");
  const ctx = canvas.getContext("2d");

  const subjects = [
    { name: "Math-I", marks: 85, color: "#0284c7" },
    { name: "BEEE", marks: 90, color: "#10b981" },
    { name: "WebTech", marks: 95, color: "#7c3aed" }
  ];

  const startX = 60;
  const barWidth = 60;
  const gap = 50;
  const baselineY = 150;

  // Draw Baseline Axis
  ctx.beginPath();
  ctx.moveTo(30, baselineY);
  ctx.lineTo(420, baselineY);
  ctx.strokeStyle = "#475569";
  ctx.lineWidth = 2;
  ctx.stroke();

  // Draw Bars & Labels
  subjects.forEach((sub, i) => {
    const x = startX + i * (barWidth + gap);
    const barHeight = sub.marks * 1.1;
    const y = baselineY - barHeight;

    // Bar Fill
    ctx.fillStyle = sub.color;
    ctx.fillRect(x, y, barWidth, barHeight);

    // Score Text
    ctx.fillStyle = "#0f172a";
    ctx.font = "bold 12px sans-serif";
    ctx.textAlign = "center";
    ctx.fillText(sub.marks + "%", x + barWidth / 2, y - 6);

    // Subject Name Text
    ctx.font = "600 11px sans-serif";
    ctx.fillStyle = "#334155";
    ctx.fillText(sub.name, x + barWidth / 2, baselineY + 18);
  });
</script>
```

🎯 Complete script renders baseline axis, scaled dynamic bars, and dual centered text annotations.